To do estimate: 267 Minutes

2026 Evil Swerve Drive To Do

File	Todo	Priority	Priority Numb	Status	Status Numb	Estimated Complete Time (Minutes)	Notes
TrajectorySystem	Make this modulo +/- PI... There is a WPILIB function for this... MathUtil.angleModulus	High	2	Open	1	5	
TurretLauncher	Need a way to set turret positon demand from joystick for testing.. Increment / Decrement value. Maybe leave this as a manual mode.	High	2	Open	1	25	LEAVE FOR LATER!!!!!….
TurretLauncher	Need a calculation of " launchertargetSpeed" For now it could be as simple as: launchertargetSpeed = 100.0; as a placeholder. Or it could be" launchertargetSpeed = 1000.0 + TurretDistance * 200.0;	High	2	Open	1	5	
TurretLauncher	Line 179 - launchertargetSpeed = 0.0; should be: useSpeedTarget = 0.0;	High	2	Open	1	2	
AgitatorSystem	If there is any motor configuration, comment out. Use Rev Hardware client to set all motor configurations.	Medium	3	Open	1	5	LEAVE FOR LATER!!!!!….
FeederSystem	If any motor configuration exists, comment out. Use Rev Hardware client to set motor configuration.	Medium	3	Open	1	5	LEAVE FOR LATER!!!!!….
IntakeSystem	Comment out the motor configuration information. Configuration can be set by Rev Hardware Client.	Medium	3	Open	1	5	LEAVE FOR LATER!!!!!….
MatchSystem	Suggest adding a "MatchSystem.java" file. This would be a pseudo static class like the other systems. This would have internal variables indicating if 4150 was Red or Blue. Add a method "disableExec" to be called when disabled to get the match info. Add "getter"functions to see if we are red or blue.	Medium	3	Open	1	15	
PhotonVision	Configure co-processor 1. Define and calibrate april tag cameras.	Medium	3	Open	1	30	May not be a programmer team task !!
PhotonVision	Configure co-processor 2. Define and calibrate april tag cameras.	Medium	3	Open	1	30	May not be a programmer team task !!
SwerveDrive	Add actual positions of modules.	Medium	3	Open	1	5	May need to measure first..
SwerveDrive	Change max speed from 13.0 to 15.0 This is a guess. This is done because the NEO Vortex motors are more powerful and have a higher maximum no load speed.	Medium	3	Open	1	5	
SwerveModule	Change max speed from 13.0 to 15.0 feet/sec. This is 4.572 meters/sec (approx). This is a guess. Change Drive_Kn = 1/4.572. (Approx line 38)	Medium	3	Open	1	5	
SwerveModule	Measured circumference of swerve wheels is 32 centimeters, or 12.5984 inches. This translates to a diameter of 4.01019527 inches. Robot currently usede 3.705. Compute a new driveDistFactor and update code. (Approx line 88)	Medium	3	Open	1	5	
SwerveTeleop	Add a driver button to execute a trajectory while button is held done. When first pressed (edge triggered) set init variable in call to follow trajectory. Will need to select trajectory based on RED/BLUE (Or just for testing….)	Medium	3	Open	1	20	SAVE FOR LATER
SwerveVision	When using single target results, make sure that ambiguity is within a reasonable range, say less than 0.3. If not then ignore value.	Medium	3	Open	1	20	SAVE FOR LATER
SwerveVision	Set actual camera names	Medium	3	Open	1	5	
SwerveVision	will have to set physical camera offset position once it is known. There is a spreadsheet in programming slack channel.	Medium	3	Open	1	15	
TrajectorySystem	add network table entries for ontarget, done, havesample, timelengthoftrajectory (rename if you don't like name)	Medium	3	Open	1	5	Some of this may be done.
TrajectorySystem	TODO: add getters for new things written to network tables. ontarget, havesample,done.	Medium	3	Open	1	5	This may be duplicate….
TurretLauncher	Use MatchSystem getter to determine red or blue field side.	Medium	3	Open	1	10	
TurretLauncher	Need to use the limit switches to stop travel. When we get a limit siwtch EDGE set the motor control position. (Make certain to back calculate rotations from degrees.)	Medium	3	Open	1	15	
TurretLauncher	Set "dump to our side" field positions Red and Blue (also maybe left and right)	Medium	3	Open	1	5	
TurretLauncher	Add LauncherSpeedDemand to network tables	Medium	3	Open	1	5	
TurretLauncher	Calculate launcher speed on target (within 75 RPM) and publish to network tables.	Medium	3	Open	1	5	

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TurretLauncher	Calculate turret on target and add value to network tables. (Should this value prevent launcher from shooting ????)	Medium	3	Open	1	10	think don't need
AgitatorSystem	Determine the best motor output to use for agitation.	Tuning	4	Open	1	?	
AgitatorSystem	Determine if speed control is needed.	Tuning	4	Open	1	?	
AgitatorSystem	Determine if the agitator should have a "reverse" mode that would be used when on and not shooting. Add a new cmd method to select the new mode, and update supervisory as needed.	Tuning	4	Open	1	?	
FeederSystem	See what output works best here. Likely should do speed control and set speed setpoint to be something less than the shooter. Maybe 100 RPM less…	Tuning	4	Open	1	?	
IntakeSystem	See what motor demand works for the intake motor.	Tuning	4	Open	1	?	
IntakeSystem	Tune the intake arm position control	Tuning	4	Open	1	?	
SwerveModule	Tune feedforward. Calculate maximum speed.	Tuning	4	Open	1	?	
SwerveModule	Tune PID. Current tuning is Kp 0.5, Ki 0, Kd 0 – all times Kn. Suggest if using integration, set a range, and max integral values to minimize windup. However ramp will likely require non zero Ki.	Tuning	4	Open	1	?	
AutoFunctions	Add AutoSpin function.	Low	5	Open	1		May not need to use. Nice to have in list of things robot can do. After GOS
AutoFunctions	Add AutoStraight function	Low	5	Open	1		May not need to use. Nice to have in list of things robot can do. After GOS
AutoSystem	Set variable AUTO_FILE_EXTENSION as final	Low	5	Open	1		
AutoSystem	In method "readFiles", variable "fileInstance" does not have an extension so it doesn't need to be removed. This doesn't cause any issues, it just isn't needed.	Low	5	Open	1		After GOS
AutoSystem	Add name of selected auto to Network Tables so drivers can verify that is being used.	Low	5	Open	1		After GOS
Climb	Do the TODO… We don't know what climb is yet.	Low	5	Open	1		Need to have build define what this is….. After GOS.
FeederSystem	Might need to do speed control here... Copy code from launcher.	Low	5	Open	1		After GOS ???
IntakeSystem	This appears to be redundant.	Low	5	Open	1		After GOS
SwerveTeleop	Remove old commented out things that will never be used.	Low	5	Open	1		After GOS
SwerveTeleop	Add driver button box ???	Low	5	Open	1		After GOS
SwerveTeleop	Add aux button box ???	Low	5	Open	1		After GOS
AgitatorSystem	Remove unused imports	Very Low	6	Open	1	?	
AgitatorSystem	Add JavaDoc	Very Low	6	Open	1		
AutoFunctions	Add JavaDoc	Very Low	6	Open	1		
AutoRoutine	Add JavaDoc	Very Low	6	Open	1		
AutoStep	Suggest adding the following Cmd -- ExecuteCmd (command name is in Function string variable)	Very Low	6	Open	1		
AutoStep	Suggest adding the following Cmd -- ExecuteCheckFor (command name is in function string variable)	Very Low	6	Open	1		
AutoStep	Suggest adding the following Cmd -- AutoDone -- This will do nothing forever....similar to wait but with a forever timeout...	Very Low	6	Open	1		
AutoStep	Add JavaDoc	Very Low	6	Open	1		
AutoSystem	Add JavaDoc	Very Low	6	Open	1		
FeederSystem	Remove unused imports.	Very Low	6	Open	1		
FeederSystem	Add JavaDoc	Very Low	6	Open	1		
IntakeSystem	Say which motor is arm and which is intake.... Is 1 intake, and 2 arm ??	Very Low	6	Open	1		
IntakeSystem	Add JavaDoc	Very Low	6	Open	1		
IntakeSystem	Is this intake or arm motor ????	Very Low	6	Open	1		
IntakeSystem	what is this for? Maybe check the actual angle -- encoderRot -- instead.	Very Low	6	Open	1		
SwerveDrive	Add JavaDoc	Very Low	6	Open	1		
SwerveDrive	Remove unused imports	Very Low	6	Open	1		
SwerveModule	Review motor configuration. Probably leave alone.	Very Low	6	Open	1		

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File	Todo	Priority	Priority Numb	Status	Status Numb	Estimated Complete Time (Minutes)	Notes
SwerveModule	Add JavaDoc	Very Low	6	Open	1		
SwerveOdometry	Add JavaDoc	Very Low	6	Open	1		
SwerveOdometry	put some comments here. For example, this is the initial robot position (pose)	Very Low	6	Open	1		
SwerveTeleop	JavaDoc	Very Low	6	Open	1		
SwerveVision	JavaDoc	Very Low	6	Open	1		
TrajectorySystem	Remove unused imports.	Very Low	6	Open	1		
TrajectorySystem	Add JavaDoc	Very Low	6	Open	1		
AutoSystem	Read of autos does NOT work. The code that tries to open the file fails. Two reasons: 1) Directory of AUTO_DIR is not used as parent, 2) File extension, AUTO_FILE_EXTENSION, is not added to name.	Very High	1	Complete	3	5	Error is caught by Try/Catch block, but no autos are read.
AutoSystem	Variable autoHashMap was not being created. Code was crashing with null exception. Add variable initialization in init function.	Very High	1	Complete	3	5	
AutoSystem	All “getters” that use variable “locExecRoutine” will break when the value of this variable is null. Need to check for not null “if (locExecRoutine != null) {“ and only get the value when this is true. Otherwise return a default value, like 0.0.	Very High	1	Complete	3	10	This causes the entire code to crash!!
AutoSystem	Does not compile. Replace with generic Exception instead of FileNotFoundException.	Very High	1	Complete	3		
SwerveModule	The NEO Vortex is NOT use spark max. I think it uses a Spark Flex controller. These are different. Need to read and convert.	Very High	1	Complete	3		
TurretLauncher	The calculate of turret angle and distance isn’t correct. The geometry methods don’t update themselves they return new objects. So Line 149 - TurretOffset.rotateBy(new Rotation2d(SwerveOdometry.getrotposition())); Needs to be TurretOffset = TurretOffset.rotateBy(new Rotation2d(SwerveOdometry.getrotposition())); Same with line 150.	Very High	1	Complete	3	5	
TurretLauncher	The variblesdesiredTurretAngleDegrees needs to be clamped to be within the limits of the turret. For now lets use -100.0, 100.0	Very High	1	Complete	3	5	
TurretLauncher	Add speed control for launcher. One PID will create output value for both motors. Add motors, Add sparkmax encoders, create average velocity, add simple motor feedforward and PID. (see swervemodule drive controller.)	Very High	1	Complete	3		Partially done. Sensors not read. Both motor velocity not averaged. Encoder object not defined in init. No need to negate. Negate is done in motor config…
TurretLauncher	Convert to degrees. Think that the default is rotations. Also take into account gear ratio...	Very High	1	Complete	3		Gear ratio is 200/20. Gear box is 16/1. Will be 4/1 soon. So for now it is 160/1, but will be 40/1.
AutoRoutine	set class variable with value of autoDescription parameter..	High	2	Complete	3		
AutoRoutine	set class variable with value of autoDescription parameter.. (2nd constructor)	High	2	Complete	3		
AutoSystem	Add “ExecuteListInit” to Robot.java. Add “ExecuteList” to robot.java.	High	2	Complete	3	5	
AutoSystem	Add code to the init function to get all available autos, then call the function to read them into the auto cache…	High	2	Complete	3		
AutoSystem	Add private static HashMap to store all trajectories <String,Trajectory<SwerveSample>	High	2	Complete	3		
AutoSystem	Add your newRoutine to the arrayList routines...	High	2	Complete	3		
AutoSystem	Add your newRoutine to the class variable to be created containing the HashMap	High	2	Complete	3		
AutoSystem	If you store the routines in the class variable containing the HashMap, then this routine doesn't need to return anything (void)	High	2	Complete	3		
AutoSystem	Add a function to return an AutoRoutine from the hashmap with a calling String parameter of the name (file name without extension)	High	2	Complete	3		
AutoSystem	Add call to TrajectorySystem.FollowTrajectory	High	2	Complete	3		Needed to add this to the individual case….

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File	Todo	Priority	Priority Numb	Status	Status Numb	Estimated Complete Time (Minutes)	Notes
IntakeSystem	scale arm position.. default is rotations.... so DEG = ROT * 360 / GEAR_RATIO. Since encoder position will start at 0 when booted, add 90.0 to the equation.	High	2	Complete	3	5	
IntakeSystem	Use built in spark max encoder. We are no longer getting a through bore encoder..	High	2	Complete	3		
IntakeSystem	Encoder definition isnt correct.. Now use spark max built in encoder .GetEncoder(); see above.	High	2	Complete	3		
IntakeSystem	limit switch is new wired to roboRIO... so this won't work. Use digital input.. Need to define object (class level), create object(in init function), read object (here).	High	2	Complete	3		
TrajectorySystem	add calculate of "complete" and return "complete" to user. complete = elapsedtime >= trajectory time length AND onTarget.	High	2	Complete	3		
TurretLauncher	Remove turret motor config. Config burned into motor from Rev Hardware Client..	High	2	Complete	3		added comments
TurretLauncher	“readSensors” is never called. Also object should be defined in init, not during each execution. Read both launcher encoders and average values. It is okay to leave launch encoder velocity in RPM.	High	2	Complete	3		
TurretLauncher	More definition of turret position control object to init from excute	High	2	Complete	3		
TurretLauncher	get the values of the goal position based on alliance. I think there is a spreadsheet in slack. For now just use RED target.	High	2	Complete	3		
TurretLauncher	This will not be throughbore encoder any longer. Use built in SparkMax encoder....	High	2	Complete	3		
TurretLauncher	Use spark max built in encoder instead. Build didn't mount through bore encoder....	High	2	Complete	3		
TurretLauncher	getPosition can be used above....	High	2	Complete	3		
AgitatorSystem	Set motor CAN Id	Medium	3	Complete	3		
AutoFunctions	Add new java file for static Auto processing functions.	Medium	3	Complete	3		
AutoFunctions	Add AutoFunctions.java file for static functions. Add an boolean AutoWait( ) function. It should set drive demand to zero and return false..	Medium	3	Complete	3		
AutoSystem	Rename method availableTrajectories to availableAutos and change references….	Medium	3	Complete	3	2	
AutoSystem	Update Robot.java to read selected auto from network tables and pass to our auto routines..	Medium	3	Complete	3		
AutoSystem	Publish list of read autos to Network Tables.	Medium	3	Complete	3		
AutoSystem	Add items to NT tables for debugging and watching autos. Suggest Step, Timeout, Parm1, Parm2, Parm3, Function. Add public getters for these.	Medium	3	Complete	3		This may be done
AutoSystem	add locNTSend trigger update.	Medium	3	Complete	3		
AutoSystem	Add call to AutoFunctions.AutoWait().	Medium	3	Complete	3		
FeederSystem	Set feeder motor CAN ID	Medium	3	Complete	3		
TurretLauncher	Set physical offset of robot to turret	Medium	3	Complete	3		Need measurement – it is approximately 6.75 from edge. Center of robot is 12.5. so offset is 5.75 Inches or 0.14605 meters
TurretLauncher	Set on / off command for launcher.	Medium	3	Complete	3		
TurretLauncher	Add limit switch values to network tables.	Medium	3	Complete	3		
TurretLauncher	TODO: Set CAN ID	Medium	3	Complete	3		
AutoRoutine	Remove unused imports	Low	5	Complete	3		
IntakeSystem	determine up and down angles on the robot -- suggest starting with 90 is up and 0 is down...	Low	5	Complete	3		
IntakeSystem	When low limit switch EDGE is true, set position to known value and send this to motor controller. Remember to reverse the scaling.	Low	5	Complete	3		
TurretLauncher	Remove unused imports	Low	5	Complete	3		
SwerveModule	Remove unused imports	Very Low	6	Complete	3		

Name	Priority
High	2
Low	5
Medium	3
Tuning	4
Very High	1
Very Low	6

Status	Priority
Open	1
Partially Complete	2
Complete	3